

第三章

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$$(1) \quad E(X) = \begin{cases} \sum_i x_i \cdot p_i, & X \text{ 是离散的} \\ \int_{-\infty}^{+\infty} x f(x) dx, & X \text{ 连续.} \end{cases}$$

(2) 令 $f(x)$ 为 X 的密度函数, $Y = g(X)$, 则

$$E(Y) = E(g(X)) = \int_{-\infty}^{+\infty} g(x) f(x) dx$$

$$(3). \quad E\left(\sum_{i=1}^n C_i X_i\right) = C_i \cdot \sum_{i=1}^n E(X_i)$$

(4) 若 X, Y 独立, 则 $E(XY) = E(X) \cdot E(Y)$

$$(5). \quad D(X) = E(X - EX)^2 = \underbrace{EX^2 - (EX)^2}_{*}$$

(1) $E(X) = \sum_{i=1}^n x_i p_i$ 点 p_i X 是离散的。它 $f(x)$, dx , X 连续。(2) 令 $f(x)$ 为 X 的密度函数, $Y = g(X)$, 则 $E(Y) = E(g(X)) = \int_{-\infty}^{+\infty} g(x) f(x) dx$ (3) $E\left(\sum_{i=1}^n C_i X_i\right) = C_i \cdot \sum_{i=1}^n E(X_i)$ (4) 若 X, Y 独立, 则 $E(XY) = E(X) \cdot E(Y)$ (5) $D(X) = E(X - EX)^2 = EX^2 - (EX)^2$ *

$$(6) \quad D(X) = \begin{cases} \sum_i (x_i - EX)^2 \cdot p_i, & X \text{ 离散} \\ \int_{-\infty}^{+\infty} (x - EX)^2 f(x) dx, & X \text{ 连续} \end{cases}$$

$$(7) \quad D(C) = 0,$$

$$D(CX) = C^2 D(X)$$

若 $X_1, X_2, \dots, X_n$ 相互独立, 则

$$D\left(\sum_{i=1}^n C_i \cdot X_i\right) = \sum_{i=1}^n C_i^2 \cdot D(X_i).$$

(8) 切比雪夫不等式

$$P(|X - EX| \geq \varepsilon) \leq \frac{D(X)}{\varepsilon^2}$$

$$\text{或 } P(|X - EX| < \varepsilon) \geq 1 - \frac{D(X)}{\varepsilon^2}$$

16) $D(X) = \sum_i (x_i - EX)^2 \cdot p_i$, X 离散 $D(X) = \int_{-\infty}^{+\infty} (x - EX)^2 f(x) dx$, X 连续
 17) $D(C) = 0$, $D(CX) = C^2 D(X)$
 18) 切比雪夫不等式 $P(|X - EX| < \varepsilon) \geq 1 - \frac{D(X)}{\varepsilon^2}$

$$\begin{aligned}
 (9) \quad \text{协方差} \quad \text{cov}(X, Y) &= E(X - EX)(Y - EY) \\
 &= E(XY) - E(X) \cdot E(Y)
 \end{aligned}$$

$$\text{相关系数} \quad \rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{DX} \cdot \sqrt{DY}}$$

$$\rho_{XY} = 0 \Leftrightarrow X, Y \text{ 不相关}$$

(10): 随机变量 X 的标准化过程:

$$X^* = \frac{X - EX}{\sqrt{DX}}$$

$$E(X^*) = 0, \quad D(X^*) = 1.$$

" 协方差 $\text{cov}(X, Y) = E[(X - EX)(Y - EY)] = E[XY] - E(X) \cdot E(Y)$ ", 相关系数 $\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{DX} \cdot \sqrt{DY}}$, " —— 厕所
 。 $\rho_{XY} = 0 \Leftrightarrow X, Y$ 不相关 , (10) : 随机变量 X 的标准化过程 : $H = \frac{X - EX}{\sqrt{DX}}$ 炒二0 , 171
 炉) = 1 ,

(11). 常见分布的期望与方差.

$X \sim (0-1)$ 分布, 则 $E(X) = p$, $D(X) = p(1-p)$.

$X \sim B(n, p)$, 则 $E(X) = np$, $D(X) = np(1-p)$.

$X \sim \pi(\lambda)$, 则 $E(X) = \lambda$, $D(X) = \lambda$

$X \sim U(a, b)$, 则 $E(X) = \frac{a+b}{2}$, $D(X) = \frac{(b-a)^2}{12}$

$X \sim \text{Exp}(\theta)$, 则 $E(X) = \theta$, $D(X) = \theta^2$

$X \sim N(\mu, \sigma^2)$, 则 $E(X) = \mu$, $D(X) = \sigma^2$

(12): 欧拉积分.

$\Gamma(\alpha) = \int_0^{+\infty} x^{\alpha-1} \cdot e^{-x} dx$ 有如下性质

$$\Gamma(n) = (n-1)!$$

$$\Gamma(\alpha+1) = \alpha \Gamma(\alpha)$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

(13) 二维随机变量 (X, Y) 的协方差矩阵

$$\Sigma = \begin{pmatrix} \text{cov}(X, X) & \text{cov}(X, Y) \\ \text{cov}(X, Y) & \text{cov}(Y, Y) \end{pmatrix}$$

$$= \begin{pmatrix} D(X) & \text{cov}(X, Y) \\ \text{cov}(X, Y) & D(Y) \end{pmatrix}$$

$$(14) \quad \text{COV}(X, Y) = E(XY) - E(X)E(Y)$$

$$E(X) = \int_{-\infty}^{+\infty} x f_X(x) dx = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x f(x, y) dx dy$$

$$E(Y) = \int_{-\infty}^{+\infty} y f_Y(y) dy = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} y f(x, y) dx dy$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f(x, y) dx dy$$

$$E(g(X, Y)) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x, y) f(x, y) dx dy$$

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$$\text{COV}(X, Y) = \rho_{XY} \cdot \sqrt{DX} \cdot \sqrt{DY}$$

$$(15) \quad \text{COV}(aX, bY) = ab \text{COV}(X, Y)$$

$$\text{COV}(X+Y, Z) = \text{COV}(X, Z) + \text{COV}(Y, Z)$$

$$D(X \pm Y) = D(X) + D(Y) \pm 2 \text{COV}(X, Y)$$

(16) 若 X, Y 独立, 则 X, Y 互不相关; 反之不成。